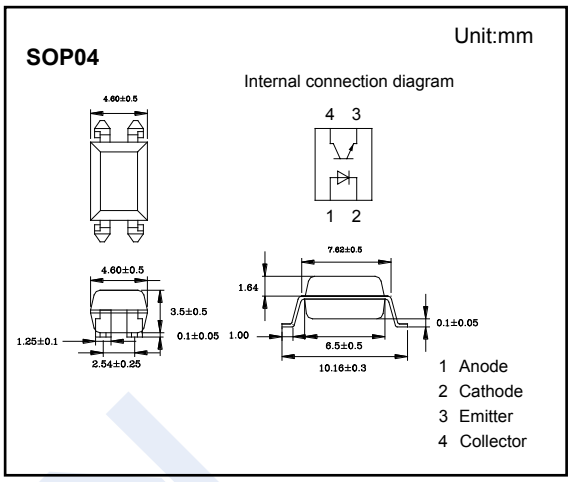
# High Density Mounting Type Photocoupler

# PC817 Series




## ■ Features

- Current transfer ratio

  ( CTR: MIN. 50% at $I_F$ = 5mA ,$V_{CE}$=5V)

- High isolation voltage between input and output

## ■ Absolute Maximum Ratings Ta = 25℃

| Parameter | Symbol | Rating | Unit |
|---|---|---|---|
| Input Reverse voltage | $V_R$ | 6 | V |
| Collector - Emitter Output Voltage | $V_{CEO}$ | 35 | |
| Emitter-Collector Output Voltage | $V_{ECO}$ | 6 | |
| Isolation Voltage | $V_{ISO}$ | 5000 | $V_{rms}$ |
| Input Forward Current | $I_F$ | 50 | mA |
| Input Peak Forward Current (Note.1) | $I_{FM}$ | 1 | A |
| Collector Current - Continuous | $I_C$ | 50 | mA |
| Input Power Dissipation | P | 70 | mW |
| Collector Output Power dissipation | Pc | 150 | |
| Total Power Dissipation | $P_{tot}$ | 200 | |
| Junction Temperature | $T_J$ | 125 | ℃ |
| Soldering temperature | $T_{sol}$ | 260 | |
| Operating Temperature | $T_{opr}$ | -30 to 100 | |
| Storage Temperature Range | $T_{stg}$ | -55 to 125 | |

Note.1:Pulse width≤100ms, Duty ratio : 0.001

# High Density Mounting Type Photocoupler

# PC817 Series

■ Electrical Characteristics Ta = 25℃

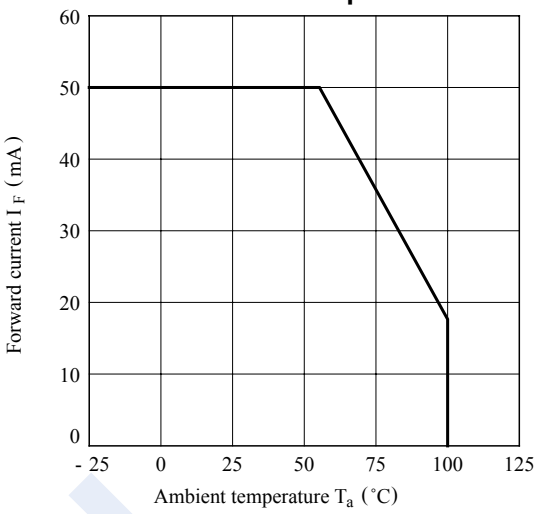
| Parameter | Symbol | Test Conditions | Min | Typ | Max | Unit |
|---|---|---|---|---|---|---|
| Input Forward Voltage | $V_F$ | $I_F$= 20 mA | | | 1.4 | V |
| Input Peak Forward Voltage | $V_{FM}$ | $I_{FM}$= 500 mA | | | 3 | |
| Input Reverse Current | $I_R$ | $V_R$= 4 V | | | 10 | uA |
| Collector- emitter cut-off current | $I_{CEO}$ | $V_{CE}$= 20 V , $I_E$= 0 | | | 10 | |
| Collector-emitter saturation voltage | $V_{CE(sat)}$ | $I_F$ = 20mA, $I_C$ = 1mA | | 0.1 | 0.2 | V |
| Isolation resistance | $R_{ISO}$ | DC 500V, 40 to 60% RH | $5 \times 10^{10}$ | $10^{11}$ | | Ω |
| Current Transfer Ratio | CTR | $V_{CE}$= 5V, $I_F$= 5mA | 50 | | 600 | % |
| Rise time | $t_r$ | $V_{CE}$ = 2V, $I_C$ = 2mA, $R_L$ = 100Ω | | 4 | 18 | uS |
| Fall time | $t_f$ | | | 3 | 18 | |
| Input Terminal Capacitance | $C_t$ | V= 0V, f=1KHz | | 30 | 250 | pF |
| Floating Capacitance | $C_f$ | | | 0.6 | 1 | |
| Cut-off frequency | $f_C$ | $V_{CE}$= 5V, $I_C$= 2mA,$R_L$=100Ω | | 80 | | KHz |

■ Classification of CTR(%)

| Type | PC817A | PC817B | PC817C | PC817D | PC817 |
|---|---|---|---|---|---|
| Range | 80-160 | 130-260 | 200-400 | 300-600 | 50-600 |

■ Typical Characterisitics

**Fig. 1 Forward Current vs. Ambient Temperature**



# High Density Mounting Type Photocoupler

# PC817 Series

■ Typical Characterisitics

**Fig. 2 Collector Power Dissipation vs. Ambient Temperature**

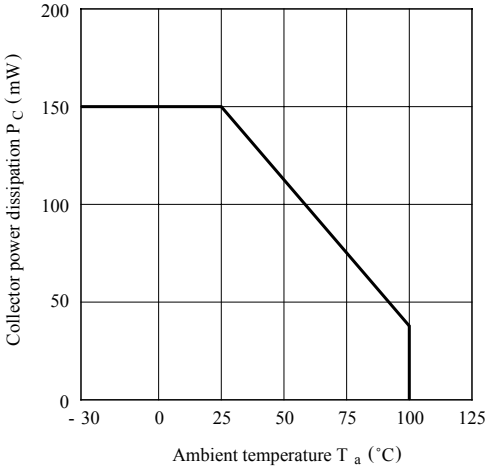



**Fig. 3 Peak Forward Current vs. Duty Ratio**

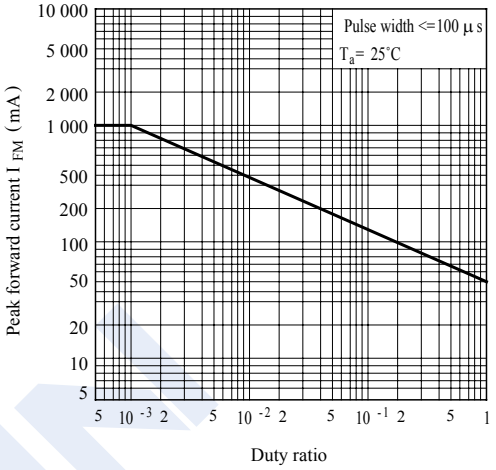



**Fig. 4 Current Transfer Ratio vs. Forward Current**

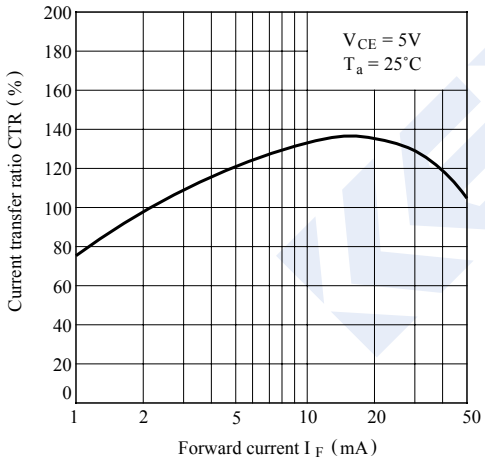



**Fig. 5 Forward Current vs. Forward Voltage**

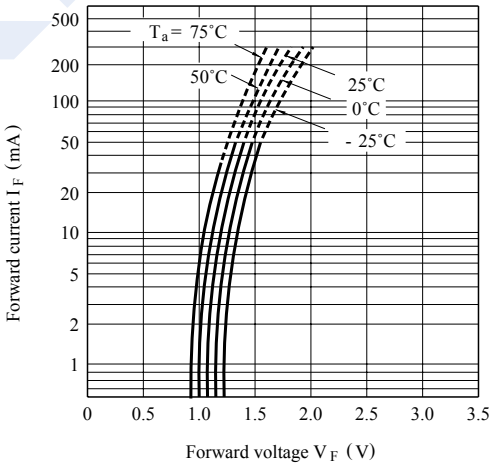



**Fig. 6 Collector Current vs. Collector-emitter Voltage**

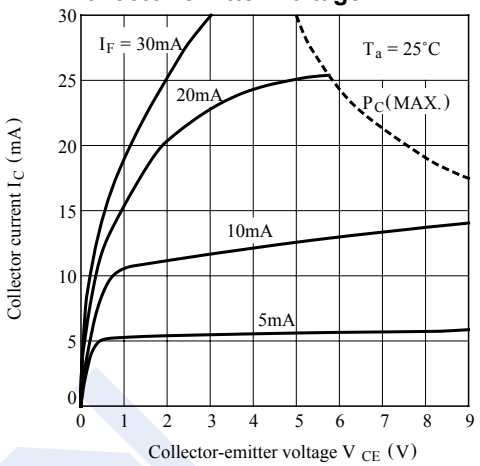
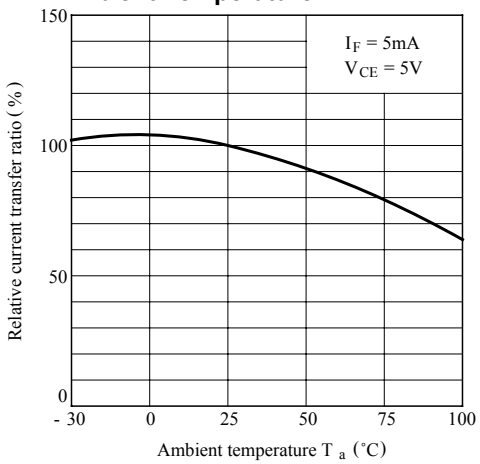



**Fig. 7 Relative Current Transfer Ratio vs. Ambient Temperature**



# High Density Mounting Type Photocoupler

# PC817 Series

■ Typical Characterisitics

**Fig. 8 Collector-emitter Saturation Voltage vs. Ambient Temperature**

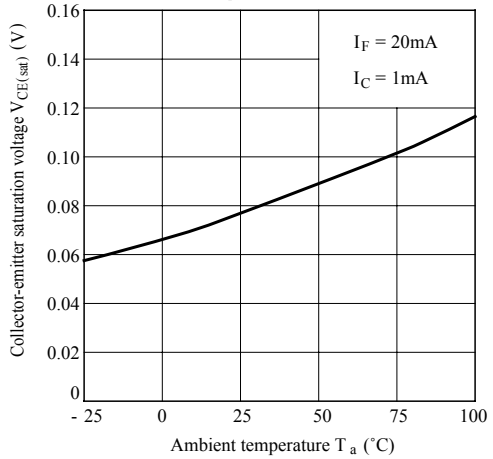



**Fig. 9 Collector Dark Current vs. Ambient Temperature**

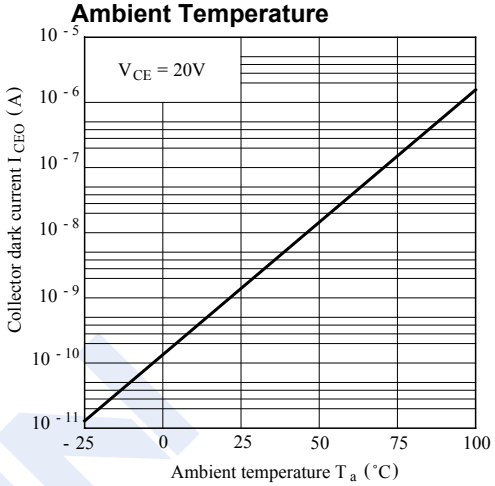



**Fig.10 Response Time vs. Load Resistance**

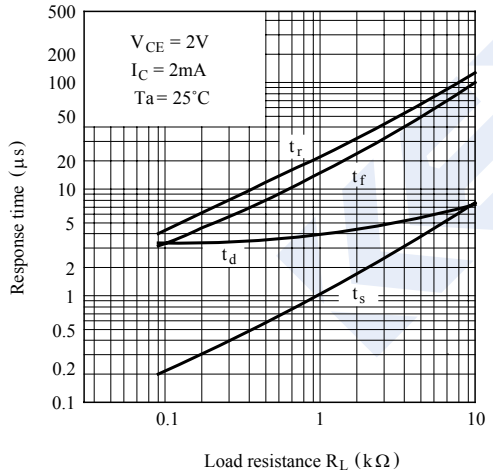
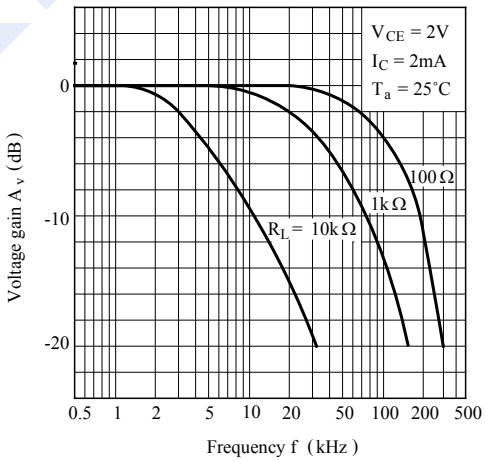
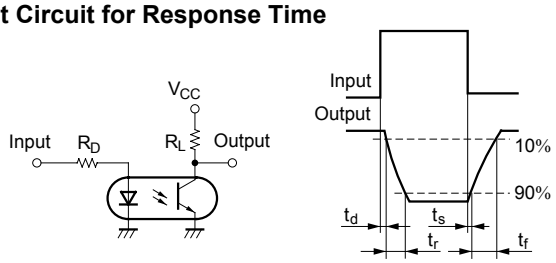



**Fig.11 Frequency Response**




**Test Circuit for Response Time**




**Test Circuit for Frequency Response**

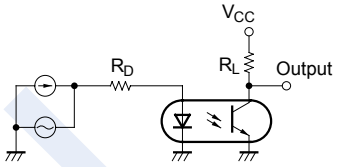



**Fig.12 Collector-emitter Saturation Voltage vs. Forward Current**

